Network Traffic Capture and Analysis Using Wireshark

1. Aim

To install and configure the Apache web server on an Ubuntu virtual machine and capture network traffic generated from a Kali Linux virtual machine using Wireshark.

The purpose of this practical is to understand and analyze different network protocols, including TCP, HTTP, DNS, and ICMP.

2. Objectives

- Install and configure Apache Web Server on Ubuntu.
- Generate network traffic from Kali Linux to Ubuntu.
- Capture TCP Three-Way Handshake packets.
- Analyze HTTP GET requests.
- Capture DNS queries and responses.
- Capture ICMP Echo Request and Echo Reply packets.
- Save the captured network traffic in PCAP format for further analysis.

3. Tools and Technologies Used

Tool	Purpose
Kali Linux	Client machine
Ubuntu Linux	Target web server
Apache2	Web server
Wireshark	Network traffic analysis
Ping	ICMP traffic generation
Curl	HTTP request generation
Nslookup	DNS query generation

4. Lab Environment

The practical was performed using two virtual machines:

- Kali Linux: Used to generate network traffic.
- Ubuntu Linux: Used to host the Apache web server.

Wireshark was used to capture and inspect the network communication between the two machines.

5. Procedure

Step 1: Installing Apache Web Server

Apache2 was installed on the Ubuntu virtual machine using:

```
sudo apt update
```

sudo apt install apache2 -y

After installation, the Apache service was started:

sudo systemctl start apache2

The service status was verified using:

sudo systemctl status apache2

Observation: Apache2 was successfully installed and its service was running.



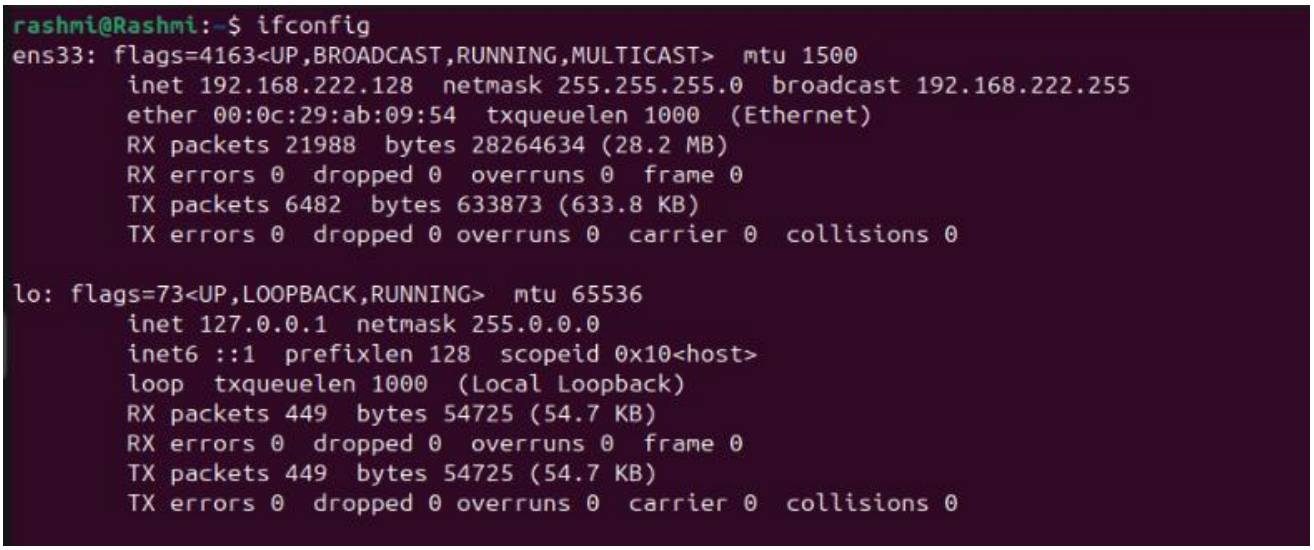
Step 2: Identifying Ubuntu IP Address

The IP address of the Ubuntu virtual machine was identified using:

ip addr

This IP address was used by Kali Linux to communicate with the Ubuntu web server.

Observation: The Ubuntu machine's IP address was successfully identified.



Step 3: Starting Wireshark Capture

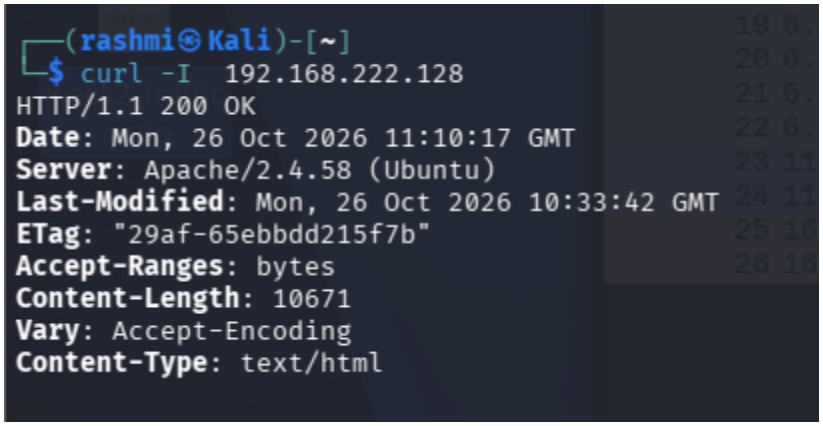
Wireshark was opened on the appropriate network interface of the Kali Linux virtual machine.

The packet capture was started before generating network traffic to ensure that all relevant packets were recorded.

Observation: Wireshark successfully captured network packets exchanged between the virtual machines.

Step 4: Capturing TCP Three-Way Handshake

An HTTP connection was initiated from Kali Linux to the Ubuntu Apache server.
The following command was used:
curl -I <Ubuntu-IP>



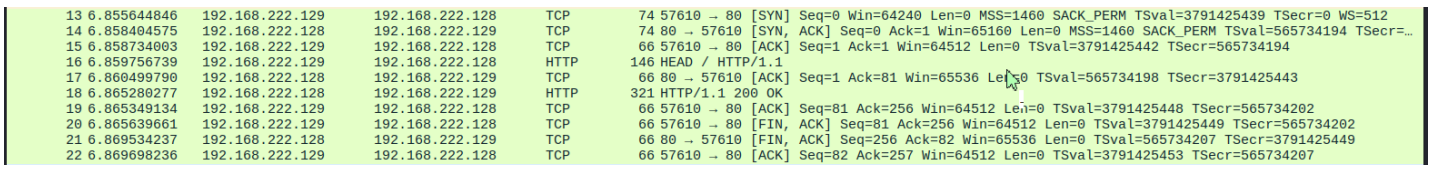
The TCP Three-Way Handshake consists of three packets:

a. **SYN**: Kali initiates a TCP connection with Ubuntu.
b. **SYN-ACK**: Ubuntu acknowledges the connection request.
c. **ACK**: Kali acknowledges the response and establishes the connection.

Wireshark display filter:

tcp.flags.syn == 1

Observation: The TCP connection establishment process was successfully captured and analyzed.



Step 6: Capturing DNS Query

The nslookup command was executed from Kali Linux to generate DNS traffic.

nslookup google.com

Wireshark display filter:

dns

Observation: DNS query and response packets were captured and analyzed in Wireshark.

23	11.138743951	192.168.222.129	192.168.222.2	DNS	82 Standard query 0x2ac1 A http://192.168.222.128
24	11.146628644	192.168.222.2	192.168.222.129	DNS	157 Standard query response 0x2ac1 No such name A http://192.168.222.128 SOA a.root-servers.net

Step 7: Capturing ICMP Echo Request and Reply

The following command was executed from Kali Linux:

ping -c 4 <Ubuntu-IP>

```
┌──(rashmi㉿Kali)-[~]
└─$ ping -c 4 192.168.222.128
PING 192.168.222.128 (192.168.222.128) 56(84) bytes of data.
64 bytes from 192.168.222.128: icmp_seq=1 ttl=64 time=9.44 ms
64 bytes from 192.168.222.128: icmp_seq=2 ttl=64 time=1.82 ms
64 bytes from 192.168.222.128: icmp_seq=3 ttl=64 time=1.52 ms
64 bytes from 192.168.222.128: icmp_seq=4 ttl=64 time=0.880 ms

── 192.168.222.128 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.880/3.415/9.440/3.494 ms
```

This generated ICMP Echo Request packets from Kali Linux to Ubuntu.

The Ubuntu machine responded with ICMP Echo Reply packets.

Wireshark display filter:

icmp

Observation: ICMP Echo Request and Echo Reply packets were successfully captured.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.222.129	192.168.222.128	ICMP	98	Echo (ping) request id=0x892b, seq=1/256, ttl=64 (reply in 2)
2	0.002291524	192.168.222.128	192.168.222.129	ICMP	98	Echo (ping) reply id=0x892b, seq=1/256, ttl=64 (request in 1)
3	1.001756998	192.168.222.129	192.168.222.128	ICMP	98	Echo (ping) request id=0x892b, seq=2/512, ttl=64 (reply in 4)
4	1.002476281	192.168.222.128	192.168.222.129	ICMP	98	Echo (ping) reply id=0x892b, seq=2/512, ttl=64 (request in 3)
5	2.026699775	192.168.222.129	192.168.222.128	ICMP	98	Echo (ping) request id=0x892b, seq=3/768, ttl=64 (reply in 6)
6	2.027628076	192.168.222.128	192.168.222.129	ICMP	98	Echo (ping) reply id=0x892b, seq=3/768, ttl=64 (request in 5)
7	3.028157200	192.168.222.129	192.168.222.128	ICMP	98	Echo (ping) request id=0x892b, seq=4/1024, ttl=64 (reply in 8)
8	3.028926754	192.168.222.128	192.168.222.129	ICMP	98	Echo (ping) reply id=0x892b, seq=4/1024, ttl=64 (request in 7)

Step 8: Saving the Captured Traffic

After completing the traffic generation activities, the packet capture was stopped and saved in PCAP format.

Filename:

Network_Traffic_Capture.pcap

The saved PCAP file can be reopened in Wireshark for further packet-level analysis.

6. Results and Observations

Protocol	Traffic Captured	Result
TCP	SYN, SYN-ACK, ACK	Successful
HTTP	GET Request	Successful
DNS	Query and Response	Successful
ICMP	Echo Request and Reply	Successful

7. Conclusion

The practical was successfully completed by installing and configuring the Apache web server on Ubuntu and generating network traffic from Kali Linux.

Wireshark was used to capture and analyze TCP Three-Way Handshake, HTTP GET requests, DNS queries, and ICMP Echo Request/Reply packets.

The captured network traffic was saved in PCAP format for future analysis. This practical provided hands-on experience in network traffic monitoring, protocol analysis, and understanding basic network communication.